Tirtha Patil

Machine Learning | Computer Vision | Explainable AI
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EDUCATION

B.Tech in Computer Science & Engineering (Artificial Intelligence & Machine Learning)
Vishwakarma Institute of Technology, Pune
CGPA: 8.22/10 | Expected Graduation: 2028
Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures, Database Management Systems, Operating Systems, Probability & Statistics, Linear Algebra

SKILLS

Programming: Python, C++, HTML/CSS, JavaScript
Machine Learning: Scikit-learn, CNN, Transfer Learning, Random Forest, SVM, SHAP, Grad-CAM
Libraries: Numpy, Pandas, Matplotlib, Seaborn, OpenCV
Tools & Frameworks: FastAPI, Git, Jupyter Notebook, VS Code

PROJECTS

AI-Powered Retina Disease Detection using CNN & Grad-CAM | GitHub
- Built MobileNetV2-based CNN achieving ~87% accuracy on retinal image dataset, implemented preprocessing, training, and validation pipeline
- Integrated Grad-CAM for explainable AI to visualize model decision regions and improve interpretability

Diabetes Outcome Prediction using Machine Learning | GitHub
- Developed ML models (Random Forest, SVM, KNN) achieving ~75% accuracy (ROC-AUC: 0.83) with preprocessing and feature scaling
- Applied SHAP for model explainability, identifying glucose and BMI as key predictive features

Medical Data Visualization & Analysis | GitHub
- Performed EDA on Pima Diabetes dataset using Pandas, NumPy, Matplotlib, Seaborn to uncover correlations between health metrics
- Identified glucose as strongest predictor and derived insights for healthcare-focused predictive modeling

Chatterbox – Real-Time WebSocket Chat Application Meeting Invite |Live Demo| GitHub
- Built real-time chat system with FastAPI and WebSockets featuring chat rooms, typing indicators, and notifications
- Designed frontend using HTML, CSS, JavaScript. Developed during Infosys Virtual Internship 6.0

EXPERIENCE

Infosys Virtual Internship 6.0 – Project Intern (Certified)
- Developed backend APIs using FastAPI.
- Implemented WebSocket-based communication.
- Collaborated in a six-member team to deliver a real-time chat application.